

1 A. BACHELOR'S DEGREE IN COMPUTER SYSTEMS ENGINEERING FROM
2 ARIZONA STATE. I HAVE BEEN BUILDING SECURE OPERATING SYSTEMS
3 FOR THE GOVERNMENT. I'VE BUILT REAL TIME TRAINERS TO TRAIN AIR
4 TRAFFIC CONTROLLERS. I TEACH COMPUTER SECURITY.

5 Q. TO WHOM DO YOU TEACH COMPUTER SECURITY?

6 A. I TEACH IT AT TECHNICAL CONFERENCES. I TAUGHT IT TO
7 GOVERNMENT AUDIENCES.

8 Q. DO YOU TEACH IN COLLEGES ALSO?

9 A. I TEACH AT THE UNIVERSITY EXTENSION.

10 Q. AND YOU HAVE -- HAVE YOU WRITTEN ARTICLES AND PRESENTED
11 TALKS ON COMPUTER SECURITY?

12 A. YES.

13 Q. ARE YOU FAMILIAR WITH AND CERTIFIED IN VARIOUS TYPES OF
14 HARDWARE, SOFTWARE, AND OPERATING SYSTEMS?

15 A. I'M FAMILIAR WITH THEM, YES.

16 Q. DO YOU USE THE COMPUTER INTERNET?

17 A. CONSTANTLY.

18 Q. FOR BOTH BUSINESS AND PERSONAL USE?

19 A. YES.

20 Q. AND APPROXIMATELY HOW LONG HAVE YOU BEEN USING THE COMPUTER
21 INTERNET?

22 A. SINCE 1983.

23 Q. ARE YOU FAMILIAR WITH HOW IT OPERATES?

24 A. YES.

25 Q. AS PART OF YOUR BUSINESS WITH DETECTING INTRUSION, DO YOU

1 NEED TO BE FAMILIAR WITH THE INTERNET?

2 A. YES.

3 Q. LET'S TALK ABOUT THE INTERNET INITIALLY. COULD YOU
4 DESCRIBE FOR US WHAT IT IS.

5 A. THE INTERNET IS A WORLDWIDE COLLECTION OF COMPUTERS TALKING
6 TO COMPUTERS, PASSING COMMUNICATIONS ACROSS VARIOUS THINGS,
7 SUCH AS LOOSE LINES, SATELLITE CHANNELS. SOME OF THOSE
8 COMPUTERS OPERATE ON BEHALF OF USERS. SOME OF THEM SERVE DATA,
9 AND SOME OF THEM SERVE TO PASS MESSAGES ACROSS THE INTERNET.

10 Q. OKAY. LET ME SHOW YOU A DIAGRAM THAT'S MARKED AS
11 GOVERNMENT'S EXHIBIT 235. ARE YOU FAMILIAR WITH THAT DIAGRAM?

12 A. YES.

13 (GOVERNMENT'S EXHIBIT NO. 235 IDENTIFIED.)

14 BY MS. MAJOR:

15 Q. AND WOULD IT AID YOU IN YOUR TESTIMONY ABOUT WHAT THE
16 INTERNET IS?

17 A. YES.

18 Q. WOULD YOU DESCRIBE FOR US IN A LITTLE MORE DETAIL WHAT THE
19 INTERNET IS AND HOW IT WORKS?

20 A. THE INTERNET IS A COLLECTION OF NETWORKS OWNED AND OPERATED
21 BY VARIOUS ORGANIZATIONS THAT COOPERATE TO PASS INFORMATION
22 ACROSS A COMPUTER NETWORK.

23 FOR EXAMPLE, IN THIS CASE WE SHOWED TWO COMPUTERS,
24 EACH OF THEM CONNECTED TO A SERVER FROM SPECIAL PURPOSE
25 COMPUTERS CALLED ROUTERS THAT PASS MESSAGES INTO THE INTERNET

1 AND ACROSS THE INTERNET TO OTHER COMPUTERS.

2 Q. AND AS -- LOOKING AT THIS DIAGRAM THERE ARE TWO LITTLE
3 CELLS, IF YOU WILL, THAT SHOW GOING FROM A COMPUTER TO AN ISP.

4 A. CORRECT.

5 Q. IN THE REAL WORLD WITH THE INTERNET, ARE THERE LITERALLY
6 MILLIONS OF THOSE POSSIBILITIES?

7 A. YES.

8 Q. AND HOW EXACTLY ARE THE COMPUTERS CONNECTED TO THE INTERNET?

9 A. THEY'RE CONNECTED OVER THE PHONE LINES, CABLE MODEMS,
10 SATELLITE CHANNELS, AND THEY COMMUNICATE BY USING STANDARD
11 PROTOCOLS. THESE ARE AGREEMENTS IN THE FORMAT OF MESSAGES TO
12 BE PASSED AROUND FROM COMPUTER TO COMPUTER.

13 Q. AND WHEN THE INTERNET WAS CREATED, IF YOU WILL, WERE THERE
14 CERTAIN RULES THAT WERE ESTABLISHED IN ORDER TO MAKE SURE THAT
15 IT WORKED?

16 A. RIGHT. AND THOSE ARE CALLED THE NETWORK PROTOCOLS.

17 Q. I'LL SHOW GOVERNMENT'S EXHIBIT 236. AND AGAIN DOES THIS
18 AID YOU IN YOUR TESTIMONY ABOUT THE PROTOCOLS?

19 A. YES.

20 (GOVERNMENT'S EXHIBIT 236 IDENTIFIED.)

21 BY MS. MAJOR:

22 Q. WOULD YOU DESCRIBE FOR US WHAT THE VARIOUS PROTOCOLS ARE
23 THAT GOVERN THE OPERATION OF THE INTERNET IN GENERAL?

24 A. STARTING AT THE BOTTOM, WE HAVE IP, THE INTERNET PROTOCOL.
25 THIS IS THE BASIC PROTOCOL THAT COMPUTERS USE TO SEND SMALL

1 PACKETS OF INFORMATION TO EACH OTHER.

2 ABOVE THAT IS A PROTOCOL CALLED TCP, TRANSMISSION
3 CONTROLLED.PROTOCOL. AND IT'S PURPOSE IS TO ESTABLISH VIRTUAL
4 CIRCUITS OR PHONE CALLS, IF YOU WILL, THROUGH THE INTERNET FROM
5 ONE COMPUTER TO ANOTHER.

6 ABOVE THAT, YOU ARE SHOWING FOUR PROTOCOLS. ONE IS
7 FOR TRANSMITTING ELECTRONIC MAIL, SMTP. ONE IS USED FOR
8 LOGGING INTO OR MAKING USE OF OTHER COMPUTERS. THAT'S TELNET.
9 IRC OR INTERNET RELAY CHAT IS A PROTOCOL THAT IS USED FOR CHAT
10 SERVICE WHEN PEOPLE WANT TO SEND REAL TIME MESSAGES TO EACH
11 OTHER IN CHAT ROOMS. AND FTP IS THE FILE TRANSFER PROTOCOL
12 WHICH IS USED FOR COPYING FILES FROM ONE COMPUTER TO ANOTHER.
13 Q. NOW, WITH REGARD TO THESE DIFFERENT PROTOCOLS, DO ALL OF
14 THE COMPUTERS CONNECTED TO THE INTERNET FOLLOW THESE SAME
15 PROTOCOLS?

16 A. YES.

17 Q. AND ARE THESE PROTOCOLS RELIABLE?

18 A. YES.

19 Q. WOULD YOU DESCRIBE FOR US HOW TCP AND IP WORK TOGETHER TO
20 ENABLE MESSAGES TO BE SENT FROM ONE COMPUTER TO ANOTHER?

21 A. THE TCP/IP PROTOCOL IS USED TO BUILD A VIRTUAL CIRCUIT,
22 LIKE A PHONE CALL THROUGH THE INTERNET. THE COMPUTERS AT EACH
23 END HAVE WHAT WE CALL INTERNET ADDRESSES OR IP ADDRESSES WHICH
24 ARE ANALOGOUS TO THE PHONE NUMBERS THAT A PERSON MIGHT HAVE.
25 AND A TCP CONNECTION OR A CONNECTION TO THE INTERNET IS

1 ESTABLISHED WHEN ONE COMPUTER MAKES A CONNECTION TO ANOTHER BY
2 KNOWING IT'S INTERNET PROTOCOL ADDRESS.

3 Q. AND WHEN THAT -- THESE TWO PROTOCOLS ARE USED TOGETHER TO
4 SEND INFORMATION FROM ONE COMPUTER TO A SPECIFIC SECOND
5 COMPUTER, DO THESE PROTOCOLS OPERATE TO MAKE SURE THAT THAT
6 MESSAGE IS SENT ACCURATELY AND TO THE CORRECT LOCATION?

7 A. YES. IF ARE YOU USING TCP AND IP TOGETHER, ITS ENTIRE
8 PURPOSE IS TO BUILD A RELIABLE CONNECTION WHERE ALL THE DATA
9 WILL EITHER BE PASSED CORRECTLY OR YOU WILL GET A -- A
10 CONNECTION WILL BE BROKEN.

11 Q. AND BASED UPON YOUR TRAINING AND EXPERIENCE DO THESE
12 PROTOCOLS OPERATE SUCCESSFULLY IN THE INTERNET?

13 A. YES.

14 Q. DESCRIBE THAT.

15 A. I'M SORRY?

16 Q. I'M SORRY. DESCRIBE THAT FOR US. HOW DO YOU KNOW THAT
17 THEY OPERATE SUCCESSFULLY?

18 A. THE TWO COMPUTERS THAT ARE COMMUNICATING WITH THE TCP
19 PROTOCOL EXCHANGE MESSAGES INDICATING THE NUMBERS OF BYTES THAT
20 HAVE BEEN TRANSMITTED. AND THEY ALSO GENERATE CHECK ZONES,
21 WHICH IS A WAY OF DETERMINING THAT A BIT OF INFORMATION HAS NOT
22 BEEN CHANGED OR LOST TO TRANSMISSION. IF THAT CHECK ZONE IS
23 INCORRECT, THEN THE CONNECTION WILL EITHER BE BROKEN OR THE
24 PACKETS WILL BE RETRANSMITTED OVER AND OVER AGAIN UNTIL THAT
25 DATA ACTUALLY MAKES IT TO ITS DESTINATION.

1 Q. SO THERE'S AN INTERNAL CHECK OR GUARANTEE, IF YOU WILL,
2 THAT WHEN THESE TWO PROTOCOLS OPERATE TOGETHER, THEY GUARANTEE
3 THAT THE INFORMATION IS TRANSMITTED ACCURATELY FROM ONE
4 COMPUTER TO ANOTHER?

5 A. YES.

6 Q. HOW DOES THE -- HOW DOES THE TCP AND IP WORKING TOGETHER,
7 HOW DO THOSE PROTOCOLS KNOW HOW TO GO FROM ONE COMPUTER TO THE
8 SECOND COMPUTER? HOW DO THEY KNOW WHICH IS THE CORRECT
9 COMPUTER TO GO TO?

10 A. THE INTERNET PROTOCOL ADDRESS IS UNIQUE WITHIN THE INTERNET
11 AT ANY TIME. THE ROUTERS -- THOSE SPECIAL PURPOSES COMPUTERS
12 THAT MAKE UP THE INTERNET -- KNOW WHEN THEY RECEIVE A PACKET OF
13 INFORMATION FOR AN IP ADDRESS, THEY KNOW WHERE TO SEND IT FOR
14 THE NEXT HOP. AND SO EVERY ROUTER PASSES IT ON TO THE NEXT HOP
15 UNTIL IT REACHES ITS FINAL DESTINATION.

16 Q. COULD YOU DESCRIBE AN INTERNET PROTOCOL ADDRESS FOR US.
17 WHAT IS THAT?

18 A. THE INTERNET PROTOCOL ADDRESS IS FOUR NUMBERS. USUALLY
19 THEY'RE PRINTED SEPARATED BY PERIODS. EACH NUMBER CAN BE
20 BETWEEN 1 AND 255. SO IT LOOKS LIKE, FOR EXAMPLE,
21 1.32.249.5.10 IS AN IP ADDRESS.

22 Q. LET ME SHOW YOU WHAT IS MARKED AS GOVERNMENT'S EXHIBIT
23 237. AND LISTED ON THIS EXHIBIT IS AN EXAMPLE OF AN IP DRESS.
24 24.0.152.23; IS THAT CORRECT?

25 A. YES.

1 (GOVERNMENT'S EXHIBIT NO. 237 IDENTIFIED.)

2 BY MS. MAJOR:

3 Q. AND IS AN INTERNET OR AN IP ADDRESS, IS AN IP ADDRESS
4 APPLIED TO ONE COMPUTER?

5 A. YES, ANY COMPUTER -- THERE SHOULD ONLY BE ONE COMPUTER
6 ATTACHED TO THE INTERNET WITH THAT IP ADDRESS AT ANY TIME.

7 Q. DOES EACH COMPUTER ALWAYS HAVE THE SAME IP ADDRESS?

8 A. THEY DON'T HAVE TO.

9 Q. EXPLAIN THAT FOR US.

10 A. COMPUTERS THAT ARE GOING TO BE USED AS SERVERS, FOR
11 EXAMPLE, WELL-KNOWN WEB SITES ITS TO THEIR ADVANTAGE TO HAVE
12 THE SAME IP ADDRESS ALL THE TIME. COMPUTERS THAT ARE ALWAYS
13 CONNECTED TO THE INTERNET, PERHAPS INSIDE THE COMPANY OR A
14 UNIVERSITY, THEY ARE EXPECTED TO HAVE THE SAME ADDRESS ALL THE
15 TIME.

16 IF YOU HAVE AN INTERNET SERVICE PROVIDER ESPECIALLY
17 ONE WHERE YOU HAVE DIAL UP ACCESS WITH A MODEM, THEY MAY HAVE
18 ONLY 50 OR 100 DIAL UP LINES, BUT THEY MAY HAVE A THOUSAND
19 CUSTOMERS. AND IN A DIAL UP ISP SITUATION, YOU MIGHT BE GIVEN
20 A NEW IP ADDRESS EVERY TIME YOU CONNECT.

21 Q. IN A DYNAMIC IP ADDRESS SITUATION, IN ORDER TO DETERMINE
22 WHOSE COMPUTER WAS USING THAT IP ADDRESS YOU NEED TO KNOW A
23 SPECIFIC TIME; IS THAT CORRECT?

24 A. CORRECT.

25 Q. WHAT ABOUT WITH A STATIC IP ADDRESS? WHAT DOES THAT MEAN?

1 A. THAT MEANS THAT THE COMPUTER ALWAYS HAS THE SAME IP
2 ADDRESS.

3 Q. HOW ARE IP ADDRESSES ASSIGNED?

4 A. THERE'S A PUBLIC TRUST ORGANIZATION CALLED THE AMERICAN
5 REGISTRY FOR INTERNET NUMBERS, AND THEY ASSIGN BLOCKS OF
6 INTERNET NUMBERS TO LARGE INTERNET SERVICE PROVIDERS, SUCH AS
7 SPRINT OR MCI OR AT HOME OR ROADRUNNER.

8 Q. AND AS YOU LOOK AT THIS IP ADDRESS, 24.0. DOES THAT MEAN
9 ANYTHING TO YOU?

10 A. THAT IS THE BLOCK OF IP ADDRESS THAT WERE ASSIGNED TO AT
11 HOME.

12 Q. AND YOU KNOW THAT FROM YOUR WORK IN THE COMPUTER INDUSTRY;
13 IS THAT TRUE?

14 A. YES.

15 Q. NOW, WHEN BLOCKS OF IP ADDRESSES ARE ASSIGNED TO THESE
16 LARGE ISP COMPANIES, WHAT HAPPENS NEXT?

17 A. A REGULAR PERSON WOULD NOT GO THE REGISTRY AND ASK FOR AN
18 IP ADDRESS. THEY WOULD HAVE TO GET AN IP ADDRESS THAT WAS
19 WITHIN THE RANGE THAT WAS ASSIGNED TO THEIR ISP. THIS IS VERY
20 SIMILAR TO BELLCORE, WHICH USED TO BE BELL LABS, WHICH ASSIGNS
21 AREAS CODES TO THE LOCAL TELEPHONE COMPANIES. SO LARGE BLOCKS
22 OF ISP'S ARE GIVEN TO COMPANIES LIKE SPRINT OR MCI THAT ARE
23 TRANSCONTINENTAL IN NATURE. AND THEN THEY CARVE OUT PIECES OF
24 THEIR ADDRESS SPACE AND THEY GIVE IT TO THE SMALLER ISP'S --
25 THE MEDIUM-SIZED ISP'S THEN TAKE THEIR ADDRESS SPACE AND ASSIGN

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1 BLOCKS OF IT TO THE SMALLEST ISP'S. AND THEN THOSE ISP'S
2 ACTUALLY ASSIGN AN INTERNET ADDRESS TO A PERSON.

3 Q. NOW, CAN MORE THAN ONE COMPUTER HAVE THE IP ADDRESS AT THE
4 SAME TIME?

5 A. YES, BUT ONLY ONE OF THEM WOULD RECEIVE THE TRAFFIC THAT
6 WAS DESTINED FOR IT.

7 Q. EXPLAIN THAT FOR US.

8 A. FOR EXAMPLE, I COULD USE THAT IP ADDRESS AND PUT IT INTO MY
9 COMPUTER AND CONNECT MY COMPUTER TO THE INTERNET, BUT BECAUSE
10 THAT IP ADDRESS WOULDN'T MATCH THE NUMBER THAT I WAS ATTACHED
11 TO, ANY MESSAGES THAT WERE SENT TO THAT IP ADDRESS WOULD BE
12 SENT TO THE COMPUTER THAT WAS ATTACHED TO THE NETWORK THAT THAT
13 ADDRESS BELONGED TO.

14 Q. SO IF YOU WERE USING AN IP ADDRESS THAT WASN'T CORRECT FOR
15 YOUR MACHINE, AND YOU EXPECTED TO GET RETURN OR A RESPONSE FROM
16 ANOTHER COMPUTER, YOU WOULD NOT RECEIVE IT?

17 A. RIGHT. THE RESPONSE WOULD GO TO THE COMPUTER THAT WAS
18 CONNECTED TO THE PROPER NETWORK.

19 Q. ARE IP ADDRESSES RELIABLE IDENTIFIERS, IF YOU WILL?

20 A. YES.

21 Q. AND WHEN THEY'RE USED IN CONNECTION -- THE IP PROTOCOL, THE
22 INTERNET PROTOCOL USED IN CONNECTION WITH THE TCP, THEY WORK
23 TOGETHER TO PROVIDE -- DO THEY WORK TOGETHER TO PROVIDE A
24 RELIABLE METHOD OF EXCHANGING INFORMATION FROM ONE COMPUTER TO
25 THE CORRECT SECOND COMPUTER?

1 A. YES.

2 Q. DOES THE INTERNET -- DOES THE FUNCTIONING OF THE INTERNET
3 RELY UPON IP NUMBERS?

4 A. YES. IF THE IP NUMBERS AND THE IP PROTOCOL DIDN'T WORK,
5 THEN THE INTERNET WOULD FALL DOWN LIKE A HOUSE OF CARDS.

6 Q. NOW, WE'VE BEEN TALKING ABOUT IP ADDRESSES AS BEING
7 NUMERICAL. IN MANY SITUATIONS, THE INFORMATION THAT'S ENTERED
8 INTO THE COMPUTER TO FIND A COMPUTER SITE ARE WORDS. HOW DOES
9 THAT WORK?

10 A. THOSE ARE CALLED DOMAIN NAMES. AND THERE IS A FUNCTION IN
11 THE INTERNET THAT TRANSLATES THOSE DOMAIN NAMES TO AN IP
12 ADDRESS, MUCH LIKE A WHITE PAGES WOULD ALLOW YOU TO FIND THE
13 PHONE NUMBER OF A PERSON.

14 Q. AND WHAT ARE THOSE SERVICES?

15 A. THOSE ARE THE -- IT'S THE DOMAIN NAME SYSTEM. AND IT WILL
16 ALLOW YOU TO MATCH A NAME TO A NUMBER OR A NUMBER TO A NAME.

17 Q. AND ARE THOSE SYSTEMS RELIABLE?

18 A. YES.

19 Q. AND DO PEOPLE IN YOUR INDUSTRY USE THAT INFORMATION
20 REGULARLY?

21 A. YES.

22 Q. WHAT ARE THE NAMES OF SOME OF THOSE SYSTEMS -- OR WHAT
23 SERVICES WOULD YOU USE? WHAT SOFTWARE WOULD YOU USE TO
24 TRANSLATE BETWEEN DOMAIN NAMES AND NUMERIC IP ADDRESSES?

25 A. THE DOMAIN NAME SYSTEMS WILL TRANSLATE EITHER WAY. THE

1 ROUTE OF THAT INFORMATION COMES FROM A REGISTRY CALLED -- IT
2 USED TO BE CALLED NETWORK INFORMATION SYSTEM, AND NOW IT'S
3 CALLED NETWORK SOLUTIONS, INCORPORATED. AND THEY CARRY THE TOP
4 LEVEL DOMAIN REGISTRATION, JUST AS AARON CARRIES THE
5 REGISTRATION OF WHO OWNS AN IP ADDRESS. NETWORK SOLUTIONS
6 CARRIES THE INFORMATION OF WHO OWNS A DOMAIN NAME, SUCH AS A
7 MICROSOFT.COM OR DOJ.GOV.

8 Q. AND IF YOU PUT -- IF YOU QUERIED THIS NETWORK SOLUTIONS
9 WITH A DOMAIN NAME, IT WOULD PROVIDE YOU WITH A CORRESPONDING
10 IP ADDRESS; IS THAT CORRECT?

11 A. IT WOULD PROVIDE ME WITH A REGISTRANT. AND FROM THAT I
12 COULD USE THE DOMAIN NAME SYSTEM TO DETERMINE THE IP ADDRESS.

13 Q. TALKING ABOUT DOMAIN NAMES, LET ME SHOW YOU WHAT IS MARKED
14 AS GOVERNMENT'S EXHIBIT 238. ARE THESE DOMAIN NAMES?

15 A. YES.

16 (GOVERNMENT'S EXHIBIT NO. 238 IDENTIFIED.)

17 BY MS. MAJOR:

18 Q. WHAT DO DOMAIN NAMES TELL YOU AS YOU LOOK AT THEM?

19 A. IN THESE CASES I CAN SEE THAT THE FIRST TWO ARE IN THE
20 DOMAIN NAME SPACE THAT'S BEEN ASSIGNED TO THE U.S. GOVERNMENT
21 BY THE ACTIVITIES OF THE ORGANIZATION PRIOR TO NETWORK
22 SOLUTIONS. THEY END "DOT GOV." AND THAT TOP LEVEL DOMAIN HAS
23 BEEN ASSIGNED TO THE U.S. GOVERNMENT.

24 MICROSOFT.COM, THIS TELLS ME THAT IT'S A DOT COM,
25 WHICH WAS ASSIGNED BY THE SAME ORGANIZATION FOR COMMERCIAL

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1 ENTITIES.

2 THE ONE THAT ENDS IN SDCA.HOME.COM, THAT IS SOME KIND
3 OF CONNECTION TO THE SAN DIEGO CALIFORNIA AT HOME INSTALLATION,
4 BECAUSE AT HOME ALWAYS TRIES TO TAG A LOCATION OF THE CABLE
5 SYSTEM AS PART OF THEIR DOMAIN NAME. ESOTERICA.PT IS A SITE
6 SOMEWHERE IN PORTUGAL.

7 Q. AND HOW DO YOU KNOW THAT?

8 A. IF IT IS NOT DOT-COM, DOT-MIL, DOT-GOV, OR DOT-ORG OR
9 DOT-EDU, THE TOP LEVEL DOMAINS, THAT'S THE PART FARTHEST TO THE
10 RIGHT, IS THE TWO-COUNTRY OR TWO-CHARACTER COUNTRY CODE. AND
11 "PT" IS THE TWO CHARACTER COUNTRY CODE FOR PORTUGAL.

12 Q. HOW ABOUT THE NEXT ONE DOWN?

13 A. THE NEXT ONE AFTER THAT IS IN ITALY BECAUSE IP IS THE
14 COUNTRY CODE FOR ITALY. THE ONE BELOW THAT IS DOT COM. MOST
15 ~~OF THE DOT COM'S ARE UNITED STATES AND ORGANIZATIONS.~~

16 AND THEN THE ONE ON THE BOTTOM I RECOGNIZE AS BEING A
17 ROUTER IN NEW YORK OPERATED BY PSI NET, WHICH IS NOW PART OF
18 UNION -- I BELIEVE PART OF UNION NET. AND THEY'RE ONE OF THOSE
19 LARGE TRANSCONTINENTAL ISP'S.

20 Q. WHEN A DOMAIN NAME HAS INFORMATION IN THERE LIKE WHITE
21 PLAINS.NEWYORK, IS THAT GENERALLY RELIABLE?

22 A. YES.

23 Q. WHAT SPECIFIC DNS SERVICES WOULD YOU USE AND RELY ON TO
24 TRANSLATE BETWEEN A DOMAIN NAME AND IP ADDRESSES?

25 A. FIRST, I WOULD USE THE DOMAIN NAME SYSTEM TO TRANSLATE THE

1 NAME TO THE IP ADDRESS. THEN I WOULD USE THE NETWORK SOLUTIONS
2 DATA BASE TO VERIFY WHO THE DOMAIN NAME BELONGED TO. AND FROM
3 THAT, I WOULD REACH ANOTHER SET OF DOMAIN NAME SERVERS TO
4 COMPARE.

5 Q. AND VERIFY THAT IT'S CORRECT?

6 A. CORRECT.

7 Q. WOULD THE SYSTEMS THAT YOU USE INCLUDE ONES LIKE TRACE
8 ROUTE, NEO TRACE, AND THE WHO IS QUERY?

9 A. THOSE ARE PROGRAMS USED FOR ESTABLISHING THE PATH THAT A
10 VIRTUAL CIRCUIT WOULD TAKE THROUGH THE ROUTERS OF THE INTERNET.
11 AND I WOULD PROBABLY ALSO USE THAT TO TRY TO DETERMINE WHICH
12 ISP WERE INVOLVED.

13 Q. AND IS THE INFORMATION THAT YOU OBTAINED ABOUT GOING
14 BETWEEN DOMAIN NAMES AND IP ADDRESSES RELIABLE?

15 A. YES.

16 Q. WHAT ABOUT THE INFORMATION OBTAINED FROM SAY TRACE ROUTE OR
17 NEO TRACE ABOUT THE PATH THAT IS TAKEN IN ORDER TO GET TO AN IP
18 ADDRESS?

19 A. YES. ALTHOUGH THE PROTOCOLS DO ALLOW THE BITS OF
20 INFORMATION TO BE SPLIT UP AND TAKE MULTIPLE PATHS THROUGH THE
21 INTERNET, RESEARCH AT U.C.S.D. AND OTHER LOCATIONS HAS
22 DETERMINED THAT ALTHOUGH POSSIBLE, IT IS MUCH LESS LIKELY THAN
23 WE EVER IMAGINED. PACKETS TEND TO TAKE THE BEST ROUTES, AND
24 THE BEST ROUTES TEND TO STAY THE BEST ROUTES FROM DAY-TO-DAY.

25 Q. IN THE INFORMATION THAT YOU OBTAINED FROM THE DOMAIN NAME

1 SERVERS AND NETWORK SOLUTION AND THE TRACE ROUTER NEO TRACE AND
2 WHO IS DATABASES, IS THAT INFORMATION OR THOSE DATA BASES,
3 DATABASES THAT PEOPLE IN YOUR INDUSTRY -- THE COMPUTER
4 INDUSTRY -- THOSE DATA BASES THAT THEY RELY ON IN GENERAL?

5 A. YES.

6 Q. AND THEY RELY ON THEM REGULARLY?

7 A. YES.

8 Q. AND THEY FOUND THEM TO BE RELIABLE?

9 A. YES.

10 Q. I'D LIKE TO TALK ABOUT FILE TRANSFER PROTOCOL. ARE YOU
11 FAMILIAR WITH THAT PROTOCOL?

12 A. YES.

13 Q. WHAT IS THAT?

14 A. THAT IS A PROTOCOL THAT IS USED FOR SENDING FILES FROM ONE
15 COMPUTER TO ANOTHER.

16 Q. AND AS WE HAD TALKED ABOUT WITH IP/TCP, IS THIS A STANDARD
17 PROTOCOL?

18 A. YES.

19 Q. IS IT ONE THAT IS USED BY COMPUTERS THROUGHOUT THE
20 INTERNET?

21 A. YES.

22 Q. WOULD YOU DESCRIBE IN GENERAL FOR US HOW IT WORKS?

23 A. ONE OF THE COMPUTERS WILL ACT AT THE FTP SERVER AND ONE OF
24 THE COMPUTERS WILL ACT AT THE FTP CLIENT. THE FTP SERVER IS
25 WHERE FILES ARE MOVED TO AND FROM. YOU USUALLY HAVE TO LOG

1 INTO AN FTP SERVER WITH A USER NAME AND PASSWORD BEFORE YOU CAN
2 SEND FILES TO IT OR DOWNLOAD FILES FROM IT, ALTHOUGH SOME FTP
3 SERVERS ARE CONFIGURED FOR WHAT'S CALLED ANONYMOUS FTP.

4 Q. LET ME SHOW WHAT IS MARKED AS GOVERNMENT'S EXHIBIT NO.
5 239. DOES THIS DIAGRAM HELP WITH YOUR TESTIMONY?

6 A. YES.

7 (GOVERNMENT'S EXHIBIT NO. 239 IDENTIFIED.)

8 BY MS. MAJOR:

9 Q. ASSUMING THAT THE SITE ON THE RIGHT -- THE BLUE SQUARE ON
10 THE RIGHT IS THE FTP SITE SERVER, THE SERVER THAT HAS THE FTP
11 SITE, AND THE ONE ON THE LEFT IS A CLIENT, WHAT DOES THAT MEAN,
12 THOSE TWO TERMS?

13 A. THE FTP SERVER IS USUALLY INSTALLED WITH THE PURPOSE OF
14 MAKING FILES AVAILABLE TO OTHERS. THAT'S THE MOST COMMON USE
15 OF IT. SOME FTP SERVERS ARE PLACES WHERE PEOPLE CAN SEND
16 FILES.

17 Q. AND THE CLIENT IN THIS PICTURE, WHAT DOES "CLIENT" MEAN?

18 A. THAT WOULD TYPICALLY BE A USER AT A PERSONAL COMPUTER.
19 THEY'RE GOING TO BE THE PERSON WHO EITHER SENDS FILES -- THE
20 FTP SERVER IS MORE LIKE A ROBOT. YOU ASKED IT FOR FILES OR YOU
21 SEND IT FILES, AND IT DOES WHATEVER ITS TOLD. THE CLIENT IS
22 USUALLY IN CONTROL OF A PERSON OR THE PERSON WOULD PUT IN THE
23 REQUESTS FOR THE FILES THAT THEY WANT TO SEND FROM THE SERVER
24 OR GET FROM THE SERVER.

25 Q. LET'S TALK ABOUT FTP OR FILE TRANSFER PROTOCOL IN THE

1 ANONYMOUS MODE FIRST. WOULD YOU DESCRIBE THAT FOR US.

2 A. AN ANONYMOUS MODE, WHICH IS ALMOST ALWAYS USED JUST TO MAKE
3 FILES AVAILABLE, THE FTP SERVER WILL ALWAYS ASK YOU FOR USER
4 NAME AND PASSWORD. IF YOU DECIDED TO HAVE AN ANONYMOUS FTP
5 SITE -- LOTS OF COMPANIES DO THIS. FOR EXAMPLE, MICROSOFT AND
6 SUN MICROSYSTEMS, CISCO, THEY MAKE SOFTWARE AVAILABLE THROUGH
7 ANONYMOUS FTP.

8 WHEN YOU HAVE AN ANONYMOUS FTP SERVER, WHEN IT ASKS
9 YOU FOR THE USER NAME, YOU TELL IT ANONYMOUS, AND THEN IT WILL
10 ACCEPT ANYTHING AS A PASSWORD. AT THAT POINT YOU CAN RETRIEVE
11 FILES FROM DESIGNATED DIRECTORIES ON THE FTP SERVER AND SEND
12 FILES TO DESIGNATED DIRECTORIES ON THE FTP SERVER.

13 Q. TAKING AS YOUR EXAMPLE OF MICROSOFT, SOMEBODY HAD TO SET UP
14 THAT MICROSOFT FTP SITE; IS THAT CORRECT?

15 A. CORRECT.

16 Q. AND IS THAT PERSON OR PERSONS -- WHATEVER, THE GROUP THAT
17 DETERMINED WHAT TYPE OF ACCESS WOULD BE POSSIBLE?

18 A. RIGHT. WHOEVER SETS UP THE FTP SITE SERVER WILL HAVE TO
19 DECIDE WHICH AREAS OF THE HARD DRIVES ARE AVAILABLE TO HAVE
20 FILES DOWNLOADED FROM OR SENT TO. THEY WILL HAVE TO CREATE THE
21 USER NAMES OR USER ACCOUNTS, AND THEY'LL HAVE TO SET THE PASS
22 WORDS FOR THOSE USER ACCOUNTS. AND THEN THEY CAN ALSO USUALLY
23 DECIDE THAT THIS USER MAY ONLY DOWNLOAD FILES OR THIS USER MAY
24 ONLY DOWNLOAD FILES FROM CERTAIN DIRECTORIES, OR THIS USER MAY
25 BE ABLE TO DOWNLOAD ALL FILES, BUT ONLY UPLOAD INTO CERTAIN

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- 1 DIRECTORIES. SO THEY HAVE COMPLETE TOTAL CONTROL OVER WHO MAY
2 RETRIEVE FILES FROM THE SERVER OR SEND FILES TO THE SERVER.
- 3 Q. AND IN THE ANONYMOUS FTP MODE, THEY GENERALLY MAKE IT SO
4 THAT ANY USER NAME WORKS WITH ANY -- OR THE ANONYMOUS USER NAME
5 WORKS WITH ANY PASSWORD; IS THAT CORRECT?
- 6 A. CORRECT.
- 7 Q. HAVE YOU USED OR ACCESSED AN FTP SITE?
- 8 A. YES.
- 9 Q. APPROXIMATELY HOW OFTEN OR HOW REGULARLY?
- 10 A. SEVERAL TIMES A DAY.
- 11 Q. HAVE YOU ALSO RUN OR CREATED AN FTP SITE?
- 12 A. YES.
- 13 Q. TELL US ABOUT THAT.
- 14 A. THE SAN DIEGO SUPERCOMPUTER CENTER AT U.C.S.D. AND OTHER
15 ORGANIZATIONS MAKE PUBLICLY AVAILABLE SOFTWARE AVAILABLE FOR
16 FREE OVER THE INTERNET. AND WE RUN ABOUT A DOZEN FTP SERVERS
17 AT THE SUPERCOMPUTER CENTER.
- 18 Q. AND AS PART OF YOUR JOB RESPONSIBILITIES DO YOU MONITOR
19 THOSE FTP SITES?
- 20 A. YES. AN IMPROPERLY CONFIGURED SITE WOULD ALLOW PEOPLE TO
21 UPLOAD FILES TO INCORRECT DIRECTORIES, AND WE'VE BEEN VERY
22 CAREFUL TO AVOID THAT.
- 23 Q. AS PART OF YOUR WORK DO YOU ALSO REVIEW THE LOGS THAT ARE
24 ASSOCIATED WITH THE FTP SITES?
- 25 A. YES.

1 Q. WHAT TYPE OF LOGS ARE THOSE?

2 A. THOSE LOGS USUALLY INDICATE WHEN A USER HAS LOGGED IN, WHAT
3 DIRECTORIES AND FOLDERS THEY HAVE VISITED, THE FILES THAT THEY
4 HAVE TRANSFERRED, EITHER DOWNLOADED FROM THE FTP SERVER OR SENT
5 TO THE FTP SERVER. THEY LOG THE IP ADDRESS OF THE OTHER
6 COMPUTER INVOLVED AND USER NAME AND PASSWORD, IF ONE WAS USED.

7 Q. IN YOUR EXPERIENCE ARE THOSE RECORDS RELIABLE?

8 A. YES.

9 Q. WHY IS THAT?

10 A. IF THEY WERE NOT -- IF THE LOGS WERE NOT RELIABLE, THERE
11 WOULD BE INCONSISTENCIES OR MISSING SECTIONS IN THE LOG.

12 Q. NOW, WE'VE BEEN TALKING INITIALLY ABOUT THESE ANONYMOUS FTP
13 SITES. IS THERE ALSO SOFTWARE THAT ENABLES INDIVIDUALS TO SET
14 UP A PRIVATE FTP SITE?

15 A. YES.

16 Q. WHAT TYPES OF SOFTWARE -- GIVE US SOME EXAMPLES.

17 A. IN THE UNIX WORLD, THE STANDARD FTP SERVER OR ONE CALLED
18 WASHINGTON UNIVERSITY FTP. IN THE PC REALM THERE'S WARFTP AND
19 SEVERAL OTHERS.

20 Q. AND DO THESE PRIVATE FTP SITES WORK ON THE SAME PRINCIPLES
21 OR PROTOCOLS AS THE ANONYMOUS FTP SITES?

22 A. YES.

23 Q. IS THERE A DIFFERENCE AS FAR AS SETTING UP ACCESS?

24 A. YES.

25 Q. AND COULD YOU DESCRIBE THAT FOR US?

1 A. IN GENERAL, EVERY FTP SERVER HAS TO HAVE A LIST OF THE
2 USERS AND THEIR NAMES AND THEIR PASSWORDS AND THEIR
3 DIRECTORIES. BUT THE FORMAT OF THOSE FILES TO CONTROL THAT
4 ACCESS WOULD CHANGE FROM OPERATING SYSTEM TO OPERATING SYSTEM
5 AND FROM FTP SERVER SOFTWARE TO FTP SERVER SOFTWARE. BUT IN
6 GENERAL, THEY HAVE TO CONTAIN THAT INFORMATION.

7 Q. NOW, IF YOU WERE RUNNING AN FTP SITE, AND I WANTED TO
8 ACCESS IT, WHAT WOULD BE THE STEPS THAT WOULD NEED TO OCCUR IN
9 ORDER FOR ME TO ACCESS IT?

10 A. FOR ANONYMOUS OR NONANONYMOUS?

11 Q. NONANONYMOUS OR PRIVATE.

12 A. FOR NONANONYMOUS FTP, I WOULD HAVE TO ACQUIRE THE SOFTWARE
13 AND INSTALL IT. I WOULD HAVE TO CREATE USER ACCOUNTS OR USER
14 NAME AND PASSWORD. I WOULD HAVE TO DECIDE WHICH DIRECTORIES
15 WOULD BE AVAILABLE TO YOU, AND WHETHER OR NOT YOU COULD
16 DOWNLOAD OR UP LOAD, AND I WOULD SOMEHOW HAVE TO GIVE YOU THE
17 USER NAME AND PASSWORD.

18 Q. WHAT ARE THE DIFFERENT WAYS OR SOME EXAMPLES OF THE WAYS IN
19 WHICH YOU COULD PROVIDE THAT USER NAME AND PASSWORD, TO ME?

20 A. I COULD CALL YOU ON THE PHONE. I COULD SEND IT IN
21 ELECTRONIC MAIL. I COULD SEND YOU A LETTER. I COULD SEND IT
22 TO YOU IN INTERNET RELAY CHAT -- JUST ANY WAY OF MAKING IT
23 AVAILABLE TO YOU.

24 Q. BUT IN SOME WAY INITIALLY THE NAME AND PASSWORD THAT YOU
25 HAVE CREATED OVER HERE AND AT YOUR SITE NEEDS TO BE GIVEN TO ME

1 IF I WERE THE CLIENT?

2 A. YES.

3 Q. ONCE I HAVE THAT USER NAME AND PASSWORD, CAN I THEN ACCESS
4 YOUR SITE?

5 A. YES.

6 Q. SO I CAN THEN USE MY COMPUTER IN THE INTERNET TO GET OVER
7 TO YOUR COMPUTER?

8 A. YES.

9 Q. ONCE I GET TO YOUR COMPUTER, ARE THERE LIMITS ON WHAT I CAN
10 DO AND WHERE CAN I GO?

11 A. THOSE LIMITS WOULD BE ESTABLISHED BY THE CONTROL FILES FOR
12 THE FTP SERVER. FOR EVERY USER THERE'S USUALLY A LIST OF
13 DIRECTORIES THAT YOU COULD DOWNLOAD FROM OR SEND INFORMATION
14 TO.

15 Q. AND THAT WOULD HAVE BEEN SET UP BY YOU AS THE FTP SITE
16 OPERATOR?

17 A. YES.

18 Q. LET'S SAY THAT IN YOUR -- WHEN YOU SET UP THIS FTP SITE,
19 YOU GAVE ME WITH MY USER NAME AND MY PASSWORD THE ABILITY TO
20 DOWNLOAD OR COPY FILES FROM YOUR SERVER, HOW WOULD THAT WORK?

21 A. ON THE CLIENT'S SIDE YOU WOULD NEED TO USE AN FTP PROGRAM
22 OR AN INTERNET BROWSER, SUCH AS MICROSOFT INTERNET EXPLORER OR
23 NETSCAPE OR ANY OF THE BROWSERS. THEY ALL SPEAK THE FTP
24 PROTOCOL.

25 YOU WOULD HAVE TO TELL THAT PROGRAM WHAT COMPUTER YOU

1 WANTED TO CONNECT TO, THEN YOU COULD LOOK AROUND AND SEE WHICH
2 DIRECTORIES WERE AVAILABLE, AND THEN YOU COULD TELL YOUR FTP
3 SOFTWARE TO DOWNLOAD FILES FROM THE FTP SERVER.

4 Q. AND WOULD I HAVE TO GO IN AND MANUALLY SAY I WANT THIS FILE,
5 I WANT THAT FILE, I WANT THE NEXT FILE?

6 A. IT DEPENDS. IF YOU WERE USING A STANDARD BROWSER OR AN FTP
7 CLIENT, YES. THERE IS ALSO SOFTWARE CALLED MIRRORING SOFTWARE.
8 AND ITS PURPOSE IS TO MAKE A COPY OF SOME PART OF THE FTP SITE
9 ON YOUR LOCAL COMPUTER.

10 Q. HAVE YOU HEARD OF A PROGRAM CALLED DELTA DIRECTORY?

11 A. YES.

12 Q. AND WHAT IS THAT?

13 A. A DELTA DIRECTORY IS A MIRRORING PROGRAM. ITS PURPOSE IS
14 TO ALLOW YOU TO HAVE YOUR COMPUTER AUTOMATICALLY DOWNLOAD AND
15 UPLOAD GROUPS OF FILES TO OR FROM ONE OR MORE FTP SERVERS.

16 Q. NOW, WHEN YOU WERE TALKING ABOUT THIS FTP, YOU SAID THAT
17 THE CLIENT THAT WANTS TO ACCESS AN FTP SITE NEEDS TO KNOW THE
18 IP ADDRESS OF THE FTP SITE; IS THAT CORRECT?

19 A. THE ACTUAL SOFTWARE DOES NEED TO KNOW THE IP ADDRESSES, BUT
20 IT WOULD BE ENOUGH FOR THE USER TO KNOW THE DOMAIN NAME.

21 Q. BUT THERE HAS TO BE EITHER A DOMAIN NAME OR AN IP ADDRESS
22 SO THAT THE CLIENT ACCESSES THE CORRECT MACHINE?

23 A. YES.

24 Q. WOULD IT BE POSSIBLE THEN FOR AN FTP SITE TO BE SET UP AT
25 SOMEBODY ELSE'S -- AT THE WRONG IP ADDRESS? DOES THAT QUESTION

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1 MAKE SENSE?

2 A. YES. THEY WOULD HAVE TO BE CONNECTED TO THE PROPER
3 NETWORK. IT WOULD BE LIKE YOU TRYING TO CLAIM THAT YOUR AREA
4 CODE ON YOUR PHONE IS 301 WHEN, IN FACT, YOU'RE CONNECTED
5 SOMEWHERE IN SAN DIEGO AND YOUR AREA CODE IS 619. IF YOUR IP
6 ADDRESS OR IF YOUR PHONE IS NOT CONNECTED TO THE RIGHT NUMBER,
7 IF YOUR COMPUTER IS NOT CONNECTED TO THE RIGHT NETWORK, THEN IT
8 JUST WOULDN'T WORK.

9 THE COURT: LET'S TAKE OUR AFTERNOON BREAK. LADIES
10 AND GENTLEMEN, YOU CAN'T SEE THE CLOCK NOW. IT'S 3 O'CLOCK.
11 IT'S ABOUT ONE MINUTE AFTER 3:00. WE'LL RESUME AGAIN AT 3:15,
12 SO WE'LL TAKE APPROXIMATELY A 15-MINUTE BREAK.

13 REMEMBER THE ADMONITION I HAVE GIVEN TO YOU PREVIOUS.

14 I ALSO WANT TO COMMENT ON A COUPLE OF THINGS. FIRST
15 OF ALL, YOU MAY RUN INTO THE LAWYERS OR STAFF OR ME IN THE
16 HALLWAY OR OVER IN THE CAFETERIA, OR PERHAPS A WITNESS THAT'S
17 INVOLVED IN THIS CASE. PLEASE DON'T STRIKE UP A CONVERSATION
18 WITH ANYONE WHO HAS ANYTHING TO DO WITH THIS CASE, AND WE'RE
19 GOING TO AVOID YOU TOO. THAT DOESN'T MEAN WE'RE UNFRIENDLY
20 PEOPLE. IT JUST MEANS THAT WE JUST DON'T WANT ANYONE TO THINK
21 THAT WE'RE TALKING ABOUT THE CASE.

22 IF YOU SEE ANYONE THAT'S INVOLVED IN THIS CASE TALKING
23 WITH ANOTHER JUROR OR ATTEMPTING TO STRIKE UP A CONVERSATION OR
24 YOU SEE A FELLOW JUROR TALKING WITH SOMEONE INVOLVED IN THE
25 CASE, PLEASE REPORT IT TO ME BY TELLING THE CLERK.

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1 MR. BRANIFF: IT WOULD BE THE WEEK OR THE WEEK AFTER.

2 MS. MAJOR: OKAY.

3 THE COURT: OKAY.

4 MR. BRANIFF: THANK YOU.

5 THE COURT: AND EXPERTS CAN STAY IN THE COURTROOM, SO
6 I WOULD ALLOW THEM TO STAY IN:

7 (RECESS TAKEN AT 3:05 P.M.)

8 (JURY PRESENT.)

9 THE COURT: YOU MAY PROCEED, MS. MAJOR.

10 MS. MAJOR: THANK YOU, YOUR HONOR.

11 BY MS. MAJOR:

12 Q. I HAVE PLACED IN FRONT OF YOU WHAT IS MARKED AS
13 GOVERNMENT'S EXHIBIT 114. DO YOU RECOGNIZE THAT?

14 A. YES.

15 (GOVERNMENT'S EXHIBIT NO. 114 IDENTIFIED.)

16 BY MS. MAJOR:

17 Q. AND WHAT IS IT?

18 A. THIS IS THE PRINTED OUTPUT FROM SOME OF THE FTP LOGS.

19 Q. HAVE YOU -- DO YOU RECOGNIZE THESE LOGS?

20 A. THEY APPEAR TO BE THE ONES THAT YOU SHOWED ME EARLIER.

21 Q. DO YOU RECOGNIZE THEM AS FTP LOGS?

22 A. YES.

23 Q. AND HAVE YOU HAD AN OPPORTUNITY TO REVIEW THEM?

24 A. YES.

25 Q. HAVE YOU REVIEWED FTP LOGS SIMILAR TO THESE BEFORE?

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1 A. YES.

2 Q. AND DO THESE LOOK LIKE OTHER FTP LOGS -- EXCUSE ME. DO
3 THEY LOOK LIKE OTHER FTP LOGS THAT YOU HAVE REVIEWED.

4 A. THE FORMAT IS DIFFERENT, BUT THEY SEEM TO CONTAIN THE SAME
5 INFORMATION.

6 Q. STARTING AT THE --

7 MS. MAJOR: I GUESS IF IT'S PROPER, CAN I ASK THE
8 COURT, CAN THE COURT SEE THAT, BECAUSE I CAN'T ASK THE JURORS?

9 THE COURT: I CAN SEE IT ON MY SMALLER ONE. ON THE
10 BIG ONE IT'S -- I CAN NOW IT'S MORE LEGIBLE.

11 JUROR: IT'S OUT OF FOCUS.

12 THE COURT: IS IT OUT OF FOCUS STILL FOR THE JURORS?
13 OKAY. SO MOST OF YOU -- IS THERE SOMEONE THAT CAN'T SEE IT
14 VERY WELL?

15 (NO RESPONSE.)

16 BY MS. MAJOR:

17 Q. MR. PERRINE, IF YOU WOULD LOOK AT THE TOP OF PAGE 1 OF THIS
18 LOG. INITIALLY WHAT DOES THAT TELL US?

19 A. THIS FIRST LOG ENTRY HAS A DATE AND TIME STAMP, AND IT
20 IDENTIFIES THE PROGRAM IT'S STARTING UP AS WAR-FTP, VERSION
21 1.66 X-4.

22 Q. AND THE DATE AND TIME STAMP THAT YOU'RE TALKING ABOUT,
23 WHERE IS THAT?

24 A. THE FAR LEFT COLUMN 1998, 05-09, 18:13.

25 Q. AND WHAT DOES THAT MEAN?

1 A. MAY 9TH, 1998, 6:13 IN THE EVENING.

2 Q. AND IS THAT BASED UPON THE TIME THAT IS SET WITHIN THE
3 COMPUTER ON WHICH THIS LOG IS RUNNING?

4 A. YES.

5 Q. COULD YOU TELL US AS YOU GO DOWN LINE BY LINE WHAT THIS LOG
6 IS SAYING.

7 A. THE SECOND LINE IS THE MESSAGE THAT COMES OUT WHEN WAR-FTP
8 IS STARTED EITHER FOR THE FIRST TIME OR AFTER THE USER DATABASE
9 HAS BEEN REMOVED. THIS IS THE USER DATABASE THAT HAS THE USER
10 NAMES, THE PASSWORDS, WHICH DIRECTORIES THE USERS ARE ALLOWED
11 TO ACCESS. AND IF THE DATABASE ISN'T THERE, IT CREATES A NEW
12 EMPTY FILE, AND IN SOME VERSIONS IT PLACES A DEFAULT USER IN
13 IT.

14 Q. AND IN CREATING A NEW FILE, WHAT DOES THAT MEAN?

15 A. IT CREATES A NEW EMPTY USER DATABASE THAT EITHER HAS NO
16 ENTRIES OR A SINGLE USER IN IT.

17 Q. AND DOES THE OWNER OF THE SITE THEN NEED TO CREATE USER
18 NAMES AND ACCESS AUTHORIZATIONS?

19 A. YES.

20 Q. DO YOU SEE THAT HAPPEN IN THIS LOG?

21 A. A LITTLE BIT FARTHER DOWN WHERE FTP IS, YOU'LL SEE THE
22 GOING OFFLINE. AND THAT WAS AT 6:17 ON THAT NIGHT. A FEW DAYS
23 LATER ON THE 12TH AT 7:51 IN THE MORNING IT STARTED AGAIN. AND
24 THAT LOOKS LIKE THAT HAPPENED WHEN THE COMPUTER WAS BOOTED MOST
25 LIKELY.

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1 THE NIGHT AFTER THAT, YOU'LL NOTICE THAT THERE'S NO
2 INDICATION THAT IT'S GOING OFFLINE. THAT PROBABLY MEANS THAT
3 THE COMPUTER WAS HALTED WITHOUT ACTUALLY SHUTTING DOWN THE FTP
4 SERVER ITSELF.

5 THEN AT 18:56, YOU'LL SEE THAT IT STARTED AGAIN AND
6 THEN AT 18:57 AND 58 USERS ARE BEGINNING TO BE CREATED.

7 Q. AND AS YOU'RE TALKING ABOUT WHERE IT'S STARTED AGAIN AND
8 THEN USERS WERE CREATED, WE'VE NOW SWITCHED TO MAY 13TH. IS
9 THAT WHAT THAT RECORD SAYS?

10 A. YES.

11 Q. AND WHERE IT SAYS, "USER "X" CREATED," AND THE NEXT LINE
12 DOWN, "USER DOCTOR CREATED," WHAT DOES THAT MEAN?

13 A. THAT MEANS THAT SOMEONE HAS USED THE WAR-FTP USER EDITOR TO
14 CREATE A USER NAMED "X" AND A USER NAMED DOCTOR. AND BELOW
15 THAT ARE THREE MORE USERS: EBI, ROG, AND PONY.

16 Q. THE NEXT TWO LINES DOWN -- FOR THOSE THAT HAVE TROUBLE
17 READING IT, IT SAYS 1998, 05-13 AT 19:05 00001, PONY, CNTR,
18 USER FROM 24.0.152.23 LOGGED IN. WHAT DOES THAT MEAN?

19 A. THAT MEANS THAT SOMEONE HAS ATTACHED TO THE FTP SERVER
20 USING THE USER NAME AND PASSWORD OF THE USER "PONY" AND THAT
21 THEY CAME FROM THE COMPUTER THAT HAD THE IP ADDRESS
22 24.0.152.23. THEN THEY WERE THE FIRST USER TO LOG ON TO THE
23 FTP SERVER AFTER THEY HAD BEEN STARTED. THAT'S THE 00001.

24 Q. THE NEXT LINE SHOWS THAT IN THE SAME SECOND, THAT THAT USER
25 LOGGED OUT. IN YOUR TRAINING AND EXPERIENCE WHAT DOES THAT

1 MEAN?

2 A. IT LOOKS LIKE THE FTP SERVER WAS BROUGHT UP. ACCOUNTS WERE
3 CREATED, AND THEN A USER -- THEN THEY LOGGED IN TO CHECK THAT
4 THE USERS HAD BEEN CREATED CORRECTLY.

5 Q. AND WHEN YOU'RE SAYING -- WHO DOES IT APPEAR FROM YOUR
6 TRAINING AND EXPERIENCE LOGGED ON TO CHECK THAT?

7 A. THEY LOGGED ON FROM THIS IP ADDRESS. AND I BELIEVE THAT'S
8 THE IP ADDRESS OF THIS COMPUTER. SO THEY LOGGED ON FROM THE
9 COMPUTER WHERE THE FTP SERVER WAS CONNECTED. THEY CONNECTED
10 BACK TO THE COMPUTER FROM ITSELF.

11 Q. AND IS THAT SOMETHING YOU SEE -- HAVE YOU SEEN THAT BEFORE?

12 A. RIGHT. THAT'S PRETTY MUCH THE FIRST STEP YOU TAKE TO MAKE
13 SURE THAT A SERVER THAT YOU ESTABLISHED IS RUNNING PROPERLY.
14 YOU JUST CONNECT TO IT FROM ITSELF.

15 Q. NOW, AS YOU LOOK AT THIS LOG -- I'M TRYING TO MAKE IT A
16 LITTLE BIT BIGGER AGAIN -- IT SHOWS A COUPLE OF LINES DOWN THAT
17 SOMEBODY'S ENTERED; IS THAT CORRECT?

18 A. YES. THAT WOULD BE USER ROG, R-O-G.

19 Q. AND FROM WHAT IP ADDRESS?

20 A. FROM IP ADDRESS 209.61.76.134.

21 Q. I'M SORRY. DID YOU SAY 134?

22 A. YES.

23 Q. AND IT'S ALSO IN FRONT OF YOU IF YOU NEED TO SEE THE RECORD
24 IF YOU ARE HAVING TROUBLE READING THIS.

25 IS THAT IP INFORMATION RELIABLE?

1 A. YES.

2 Q. WHY?

3 A. BECAUSE IF THAT IP ADDRESS WAS NOT CORRECT, THEN WHEN THE
4 FTP SERVER USING THE TCP PROTOCOL TRIED TO SEND THE DATA OR
5 ACTUALLY FINISH MAKING THE CONNECTION FROM THE CLIENT TO THE
6 SERVER, THE RESPONSES WOULD HAVE GONE BACK TO ANOTHER COMPUTER.

7 Q. THERE'S A NUMBER THAT WE HAD TALKED ABOUT BEFORE WHEN YOU
8 WERE TALKING ABOUT PONY, RIGHT BEFORE PONY IT SAID 00001. AND
9 IN FRONT OF ROG IT SAYS 00002. WHAT DOES THAT NUMBER REFER TO?

10 A. THAT'S A SESSION NUMBER. AND EVERY TIME A USER LOGS IN,
11 THAT CREATES A SESSION. THE SESSION NUMBER IS INCREMENTED
12 EVERY TIME A NEW USER LOGS IN.

13 Q. AND WOULD THAT -- DOES THE PERSON WHO HAS ACCESSED THE
14 SITE MAINTAIN OR KEEP THAT SAME SESSION NUMBER THE ENTIRE TIME
15 HE OR SHE IS ON?

16 A. THEY KEEP IT DURING ALL THEIR FILE TRANSFERS UNTIL THEY
17 LOG OFF OR ARE DISCONNECTED.

18 Q. IS THERE A WAY TO TELL WHETHER A USER HAS LOGGED ON
19 SUCCESSFULLY?

20 A. IF THE USER LOGS IN SUCCESSFULLY, YOU'LL SEE THE MESSAGE
21 LOGGED IN, AND THERE'S A DIFFERENT MESSAGE THAT SAYS THAT THE
22 USER WAS NOT ABLE TO LOG IN CORRECTLY.

23 Q. MOVING ON DOWN, DO YOU SEE AN EXAMPLE OF SOMEONE WHO WAS
24 UNABLE TO LOG IN?

25 A. IT'S USER JJ. IT'S SESSION THREE. USER JJ, WRONG

1 | PASSWORD, LOG IN REFUSED. SO THEY HAVE THE USER NAME CORRECT,
2 | BUT THEY DID NOT KNOW THE PASSWORD. AND THAT USER CAME FROM IP
3 | ADDRESS 24.64.154.130 BECAUSE THE SESSION NUMBERS MATCH IN
4 | THOSE TWO LOG ENTRIES.

5 | Q. AND SO WHERE IT SAYS THAT THE WRONG PASSWORD WAS USED, IT
6 | SHOWS THAT THE USER COMING FROM THAT IP ADDRESS HAD A CORRECT
7 | USER NAME, BUT THE WRONG PASSWORD; IS THAT RIGHT?

8 | A. CORRECT.

9 | Q. AND ARE THOSE RECORDS ACCURATE?

10 | A. YES.

11 | Q. OR RELIABLE?

12 | LET ME ASK YOU. IS IT POSSIBLE FOR MORE THAN ONE PERSON TO
13 | USE THE SAME USER NAME AND PASSWORD?

14 | A. YES.

15 | Q. AND HOW WOULD THAT WORK?

16 | A. THE WAR-FTP SOFTWARE IN THIS VERSION HAS A SWITCH THAT
17 | DETERMINES WHETHER A USER IS ALLOWED TO LOG ON MORE THAN ONCE
18 | AT THE SAME TIME. IF THEY DO, IF TWO PEOPLE AT DIFFERENT
19 | COMPUTERS KNEW THE USER NAME AND THE PASSWORD, THEY COULD BOTH
20 | LOG IN AT THE SAME TIME, BUT THEIR SESSION NUMBERS WOULD BE
21 | DIFFERENT.

22 | Q. SO WHEN YOU COMPARE THE SESSION NUMBER WITH THE NAME AND
23 | THE IP ADDRESS, YOU CAN DETERMINE WHICH PERSON HAS LOGGED IN;
24 | IS THAT CORRECT?

25 | A. WHICH PERSON HAS LOGGED IN, THE COMPUTER THAT IT CAME FROM,

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1 AND WHICH FILES THEY EITHER UPLOADED OR DOWNLOADED.

2 Q. LET ME SHOW YOU ANOTHER EXAMPLE. IT'S ON PAGE 10. TOWARDS
3 THE TOP OF THIS DOCUMENT AT MAY 13TH AT THE TIME OF 1952, DO
4 YOU SEE A USER USING Z-I-P AS A NAME ATTEMPT TO ENTER?

5 A. YES.

6 Q. TELL US WHAT THAT INFORMATION SHOWS?

7 A. THAT LOG RECORD SHOWS THE TIME THAT A USER ATTEMPTED TO LOG
8 IN USING THE USER NAME ZIP, AND THE SYSTEM HAD NOT BEEN
9 CONFIGURED TO NOTE THAT AS A USER NAME. AND THE LINE FOLLOWING
10 THAT SHOWS THE IP ADDRESS THAT THEY CAME FROM, WHICH IS
11 209.166.133.55.

12 Q. AS YOU CONTINUE DOWN THIS PAGE OF THE LOG, WHAT DOES IT
13 SHOW YOU ABOUT WHAT -- WHAT DOES IT TELL YOU ABOUT WHAT ZIP IS
14 DOING?

15 A. ZIP TRIES TO LOG IN SEVERAL TIMES, BUT THAT ACCOUNT DOESN'T
16 SEEM TO BE -- IT IS NOT CONFIGURED IN THE FTP SERVER.

17 Q. FARTHER DOWN APPROXIMATELY IN THE MIDDLE OF THE CHART AS
18 IT'S NOW SHOWN TO YOU, DO YOU SEE WHERE IT SAYS USER Z-I-P-P
19 CREATED?

20 A. YES.

21 Q. WHAT DOES THAT INDICATE?

22 A. THAT INDICATES THAT SOMEONE AT THE COMPUTER OPENED THE
23 WAR-FTP USER EDITOR AND CREATED AN ACCOUNT CALLED Z-I-P-P.

24 Q. ON DOWN A COUPLE OF LINES FURTHER DOWN DO YOU SEE WHETHER
25 AN INDIVIDUAL USING THE USER NAME Z-I-P-P ENTERS?

1 A. YES.

2 Q. AND WHERE IS THAT?

3 A. THEY CAME IN THERE WAS SESSION NO. 18. THEY CAME FROM IP
4 ADDRESS 209.166.133.55, WHICH IS THE SAME IP ADDRESS THAT WAS
5 USED BY -- OR ATTEMPTED TO USE THE USER Z-I-P.

6 Q. AND NOW THEY ARE ABLE TO ENTER AS Z-I-P-P?

7 A. RIGHT.

8 Q. GOING BACK TO PAGE 1 DOWN TOWARDS THE BOTTOM, LOOKING AT
9 02ROG, WHAT DOES THAT INFORMATION TELL YOU?

10 A. IS THAT THE LINE AT 1914?

11 Q. YES.

12 A. THIS SHOWS THAT USER ROG DOWNLOADED A FILE FROM THE "C"
13 DRIVE IN THE FOLDERS BINARIES, YOUNG PICTURES B2HMN, BOYS,
14 ANAL, ACTION, B2HMN04.JPG, WHICH IS AN IMAGE FILE.

15 Q. AND WHAT DOES IT SAY THERE AFTER THAT?

16 A. IT TELLS THE SIZE OF THE FILE OR THE SIZE OF THE -- THE
17 AMOUNT OF DATA THAT WAS TRANSFERRED WHICH WAS 6,052 BYTES, AND
18 IT ALSO TELLS THE SPEED OF THE CONNECTION WHICH WAS 5.9
19 KILOBYTES PER SECOND.

20 Q. AND WHAT IS THAT INFORMATION, THE SIZE AND THE SPEED OF THE
21 KILOBYTES PER SECOND, WHAT IS THAT TELLING -- OR WHAT IS THAT
22 INFORMATION?

23 A. THIS INDICATES THAT A FILE OF THAT SIZE WAS SUCCESSFULLY
24 TRANSMITTED TO THE FTP SERVER OR FROM THE SERVER TO THE CLIENT
25 AND ABOUT HOW FAST THAT FILE WAS TRANSMITTED.

1 Q. AND HOW DO YOU KNOW THAT IT WAS SUCCESSFULLY SENT?

2 A. THERE'S A DIFFERENT MESSAGE THAT IS DISPLAYED IF THE FILE
3 TRANSFER IS INTERRUPTED FOR ANY REASON.

4 Q. ON THAT LINE IT SAYS, "FILE SENT SUCCESSFULLY." HOW DOES
5 THE FTP PROGRAM KNOW THAT?

6 A. IF THE FILE TRANSFER WAS INTERRUPTED, THE PROTOCOL WOULD
7 HAVE FILED AND THE FTP SERVER WOULD HAVE NOTICED THE ERROR
8 CONDITION AND USED -- AND LOGGED A DIFFERENT ERROR MESSAGE.

9 TO BE SUCCESSFUL, ALL THE DATA HAS TO BE SENT FROM
10 THE SERVER TO THE CLIENT OR VICE VERSA, AND THEN BOTH ENDS
11 AGREE THAT THE DATA WAS TRANSMITTED COMPLETELY, AND THEN AT
12 THAT POINT THEY WOULD BREAK THE CONNECTION AND GET A SUCCESSFUL
13 LOG-IN SESSION.

14 Q. AND DO THE TWO COMPUTERS, THE COMPUTER AT THE FTP SITE AND
15 THE CLIENT COMPUTER NEED TO COMMUNICATE TO VERIFY THAT THE
16 COMMUNICATION WAS SUCCESSFUL?

17 A. RIGHT. THIS TAKES A COMPLETE TCP/IP VIRTUAL CIRCUIT PHONE
18 CALL KIND OF CONNECTION.

19 Q. SO THE IP ADDRESS NEEDS TO BE RELIABLE?

20 A. YES.

21 Q. STILL LOOKING AT ROG, THERE'S A SECTION HERE WHERE IT TALKS
22 ABOUT ANOTHER PICTURE AND IT SAYS "SENDING." WHAT DOES THAT
23 TELL YOU?

24 A. THIS IS A LOG MESSAGE THAT IS USED BY WAR-FTP TO TELL YOU
25 IT HAS BEGUN SENDING THE DATA FILE, BECAUSE THE TRANSFER COULD

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1 TAKE SECONDS, MINUTES, OR EVEN HOURS. THERE'S A LOG MESSAGE
2 THAT'S ENTERED WHEN THE FILE TRANSFER BEGINS, AND THEN THERE'S
3 A LOG MESSAGE THAT IS ENTERED WHEN THE FILE TRANSFER IS
4 FINISHED.

5 Q. LOOKING AT THE TWO MESSAGES WE WERE JUST LOOKING AT WHERE
6 IT LISTS A PICTURE BY ITS ENTIRE PATH SITE AND THE FACT THAT
7 IT'S SENDING, THEN THE SECOND MESSAGE AGAIN LISTING THE SAME
8 PICTURE BY THE SAME PATH SITE AND SAYING THAT THE FILE WAS SENT
9 SUCCESSFULLY, WHAT DOES THAT TELL YOU ABOUT WHERE THAT PICTURE
10 WAS SENT?

11 A. THAT PICTURE WAS SENT TO THE COMPUTER THAT MATCHES SESSION
12 00002. AND TO FIND THAT I WOULD LOOK IN THE LOG RECORD FOR THE
13 LOG-IN FOR ROG SESSION 00002.

14 Q. AND SO THE PICTURE WAS SENT TO THE IP ADDRESS THAT ROG
15 ENTERED FROM IN SESSION TWO?

16 A. YES.

17 Q. LET ME SHOW YOU ON PAGE 9 WE TALKED ABOUT TRANSFERS THAT
18 WERE NOT SUCCESSFUL. LOOKING WHERE MY FINGER IS, IT'S ON PAGE
19 9, ABOUT AN INCH AND A HALF UP FROM THE BOTTOM. THERE'S A --
20 DO YOU SEE AN ENTRY AT MAY 13TH AT 1951 INVOLVING JJ AND
21 SESSION 11?

22 A. YES.

23 Q. TELL US WHAT YOU SEE THERE.

24 A. IN THIS CASE THE TRANSFER HAD BEGUN, BUT WAS INTERRUPTED
25 WHEN AN ABORT COMMAND WAS RECEIVED FROM THE CLIENT SOFTWARE.

1 THIS IS THE EQUIVALENT TO HITTING A STOP IN A BROWSER WHEN YOU
2 ARE IN THE MIDDLE OF LOADING A WEB PAGE, AND YOU'VE DECIDING
3 THAT YOU DON'T WANT TO WAIT OR YOU'RE OUT OF DISK SPACE OR YOU
4 JUST WANT TO STOP A TRANSFER. THE CLIENT SENDS AN ABORT
5 MESSAGE TO A SERVER AND THE SERVER RECEIVES THAT AND THEN STOPS
6 SENDING THE DATA AND THEN GENERATES THIS LOG MESSAGE.

7 Q. ARE THERE ALSO SITUATIONS WHERE FOR SOME REASON OTHER THAN
8 THE FACT THAT THE CLIENT STOPPED IT, THAT THE TRANSMISSION WAS
9 NOT SUCCESSFUL?

10 A. YES. IF A ROUTER -- IF A LARGE NUMBER OF ROUTERS ALONG
11 THE WAY OR A ROUTER AT ONE OF THE ISP'S OR SOMETHING HAD CAUSED
12 ONE OF THE COMPUTERS TO BE DISCONNECTED FROM THE INTERNET, THEN
13 THERE'S A DIFFERENT ERROR MESSAGE WHICH COMES OUT.

14 Q. SHOWING YOU ON PAGE 71 TOWARDS THE TOP, DO YOU SEE AN
15 EXAMPLE OF THE TYPE OF TRANSACTION THAT YOU WERE JUST TALKING
16 ABOUT?

17 A. USER PONY, SESSION 21 AT MAY 24TH AT 2314, WHICH WOULD BE
18 11:14 P.M., DURING THE TRANSFER OF THE FILE ANTHON.1-AVI SOCKET
19 ERROR. TRANSFER ABORTED. THAT INDICATES SOME KIND OF ERROR IN
20 THE NETWORK OR AT ONE OF THE OTHER OF THE ENDS OF THE SESSION.

21 Q. AND THAT ALERTS THE INDIVIDUAL THAT THIS TRANSFER WAS NOT
22 SUCCESSFUL?

23 A. CORRECT. THIS IS THE FTP SERVER SITE SOFTWARE DETECTING
24 THAT THE FILE WAS NOT SENT CORRECTLY DUE TO SOME NETWORK
25 PROBLEM.

1 Q. WHEN A FILE LIKE THIS ONE ENDS IN AVI, WHAT DOES THAT MEAN?
2 A. I BELIEVE THAT'S A MOVIE.
3 Q. NOW, IS IT POSSIBLE FOR THE USER TO DECIDE TO TURN THE SITE
4 OFF?
5 A. THE CLIENT OR THE SERVER?
6 Q. THE SERVER.
7 A. YES.
8 Q. ON PAGE 55 DO YOU SEE AN EXAMPLE OF THAT?
9 A. AT MAY 14TH 8:00 A.M. THERE'S A MESSAGE THAT SAYS, "GOING
10 OFFLINE. ANY CONNECTION WILL BE CLOSED." THIS INDICATES THAT
11 THE USER AT THE COMPUTER HAS DECIDED TO TURN OFF THE FTP
12 SERVER.
13 Q. AND SO CAN ANYONE ACCESS THE SITE NOW?
14 A. THEY CANNOT ACCESS THE FTP SERVER.
15 Q. ARE THERE ALSO PARAMETERS THAT CAN BE SET REGARDING IDLE
16 TIME?
17 A. YES. YOU CAN ALSO DETERMINE THAT IF A USER HAS NOT
18 DOWNLOADED A FILE OR UPLOADED A FILE OR OTHERWISE HASN'T
19 INTERACTED WITH A SERVER FOR SOME PERIOD OF TIME, THEY WILL
20 AUTOMATICALLY BE LOGGED OUT TO MAKE ROOM FOR OTHER USERS TO
21 CONNECT.
22 Q. IF YOU WOULD LOOK PLEASE AT PAGES 72 THROUGH 75. WHAT DO
23 YOU SEE ON THESE PAGES AND WHAT DO THEY TELL YOU?
24 A. ON THE 24TH AT 2355, THE SERVER WAS TURNED OFF. ON THE
25 25TH AT 1752 THE FTP SERVER WAS STARTED AGAIN. AND AT THIS

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PERRINE - DIRECT EXAM BY MS. MAJOR, 1-12-00

1 POINT IT WAS UNABLE TO OPEN THE USER DATABASE. THE USER
2 DATABASE WAS NOT THERE OR COULDN'T BE FOUND, AND SO THE SERVER
3 AUTOMATICALLY CREATED A NEW FILE AND APPARENTLY ADDED A SAMPLE
4 USER.

5 AND THEN SHORTLY AFTER THIS THERE ARE A LARGE NUMBER
6 OF ATTEMPTS TO LOG INTO THE FTP SERVER FROM THE COMPUTER ITSELF
7 USING A LARGE NUMBER OF DIFFERENT USER NAMES SUCH AS "A" AND
8 "ABC" AND OTHERS. AND THE FTP SERVER STARTED AND RESTARTED A
9 FEW TIMES AS WELL.

10 AND THEN FINALLY ON PAGE 75 AT 1809 A USER AAAA IS
11 CREATED BY SOMEONE RUNNING THE USER EDITOR AT THE COMPUTER AND
12 THEN THEY'RE ABLE TO LOG IN AND TEST THE SERVER FROM THE
13 COMPUTER ITSELF. AND THEN BELOW THAT THE USER JEANDOUX,
14 J-E-A-N-D-O-U-X IS CREATED AND THE USER "X" IS CREATED.

15 Q. AND GOING UP TO THE PART YOU WERE TALKING ABOUT WHERE
16 THERE'S A BUNCH OF ILLEGAL USER I.D.'S AND UNSUCCESSFUL
17 ATTEMPTS TO ENTER, THOSE WERE ALL FROM THE IP ADDRESS
18 24.0.152.23; IS THAT CORRECT?

19 A. YES.

20 Q. AND FROM YOUR REVIEW OF THESE FILES WHAT DOES THAT TELL YOU
21 ABOUT THAT ADDRESS? IS THAT WHERE THE FTP SITE IS LOCATED?

22 A. YES.

23 Q. BASED UPON YOUR TRAINING AND EXPERIENCE, WOULD IT HAVE
24 BEEN -- I GUESS WE CAN TURN THIS OFF.

25 BASED UPON YOUR TRAINING AND EXPERIENCE IN YOUR REVIEW

1 OF THESE RECO
2 HAVE SET UP
3 LOCATION?

4 A. IT IS --

5 Q. WHAT WOULD

6 A. THEY WOULD

7 FIND SOME EXP

8 DESIGN OF OPERA

9 FOUND IN IT I

10 THEY WOULD BE

11 COMPUTER. TH

12 SOFTWARE IN C

13 HAVE TO INSTA

14 THE FTP SERV

15 IMAGES.

16 Q. HOW MUCH

17 GENERALLY TAK

18 A. AN AWFUL

19 Q. WHAT ABOUT

20 A. A MOVIE F

21 Q. IF SOMEB

22 YOU JUST OUTD

23 COMPUTER WHA

24 USE THEIR COM

25 A. THEY WOULD

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PERRINE - CROSS-EXAM BY MR. BRANIFF, 1-12-00

1 MS. MASOR: I HAVE NO FURTHER QUESTIONS.

2 THE COURT: MR. BRANIFF?

3 MR. BRANIFF: THANK YOU, YOUR HONOR.

4 CROSS-EXAMINATION

5 BY MR. BRANIFF:

6 Q NOW, MR. PERRINE, YOU'VE TOLD US THAT YOU WORK FOR SAN
7 OF SUPERCOMPUTER. IS THAT ONE COMPUTER OR MANY COMPUTERS?

8 A WE HAVE, DEPENDING ON HOW YOU COUNT, BETWEEN 300 AND 1500
9 COMPUTERS TO DATE.

10 Q AND ARE THEY ALL CONNECTED IN SOME WAY OR ARE MANY OF THEM
11 CONNECTED?

12 A THEY ARE ALL CONNECTED.

13 Q THEY'RE ALL CONNECTED TO EACH OTHER?

14 A THEY ARE ALL CONNECTED TO THE INTERNET AND, THEREFORE, TO
15 EACH OTHER.

16 Q NOW, IS THERE A "SUPERCOMPUTER" OR IS THAT THE COMPOSITE
17 GROUP NUMBER?

18 A WE HAVE SEVERAL LARGE MACHINES THAT ARE CLASSIFIED AS
19 SUPERCOMPUTERS.

20 Q YOU INDICATED THAT ATTEMPTS ARE MADE TO GET INTO YOUR SYSTEM
21 ABOUT 200 TIMES A WEEK; IS THAT CORRECT?

22 A YES.

23 Q AND WHEN YOU SAY "THE SYSTEM," DO YOU MEAN ONE OF THESE
24 SMALL COMPUTERS OR THE BIG COMPUTERS OR --

25 A IN AGGREGATE WE HAVE SEVERAL HUNDRED DESKTOP COMPUTERS, AS